



B.Tech. Degree IV Semester Examination April 2017

ME 15-1403 MECHATRONICS (2015 Scheme)

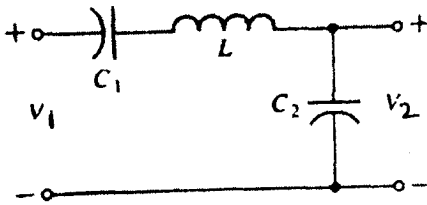
Time: 3 Hours

Maximum Marks: 60

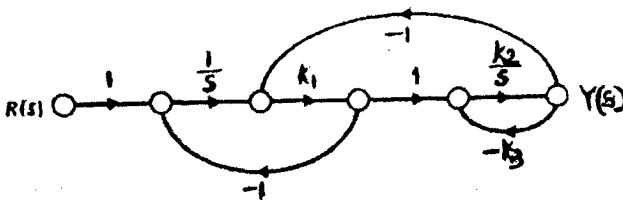
PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) Describe how proximity of an object can be detected using a pneumatic sensor.
 (b) What will be the change in resistance of an electrical resistance strain gauge with a gauge factor of 2.1 and resistance 50 Ω if it is subjected to a strain of 0.001?
 (c) Explain the working of a vane pump.
 (d) Describe the basic building blocks for modeling thermal systems.
 (e) A switching circuit is used to convert one level of dc voltage to an output dc voltage. The filter circuit to filter out the high frequencies is shown in the figure. Calculate the transfer function $V_2(s)/V_1(s)$.



- (f) Determine the closed loop transfer function $T(s) = (Y(s))/(R(s))$ for the system shown in figure.



- (g) Determine the stability of the system represented by the characteristic equation $s^4 + 9s^3 + 45s^2 + 87s + 50 = 0$ using Routh-Hurwitz criteria.
 (h) A robot arm and camera is used to pick fruits. The camera is used to close the feedback loop to a microcomputer, which controls the arm. The process is $G(s) = \frac{K}{(s+3)^2}$. The system is shown by the signal flow graph given below.



Calculate the expected steady-state error of the gripper for a unit step input as a function of K.

- (i) How many steps/pulses per second will a microprocessor need to output per second to a stepper motor if the motor is to give an output of 0.25 rev/s and has a step angle of 7.5° ?
- (j) Differentiate a microprocessor and a microcontroller.

PART B

(4 × 10 = 40)

- II. (a) Describe the working of a 3-bit absolute encoder, using neat diagrams. How many angular positions can an encoder with 8 tracks specify? (5)
- (b) Design a pneumatic circuit to sequence two cylinders in the following order, cylinder A extends; then cylinder B extends. Use suitable direction control valves, limit switches, relays etc. Explain the circuit. (5)

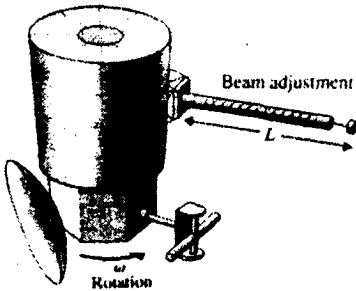
OR

- III. (a) Explain: (i) piezoelectric sensor (ii) LVDT. (4)
- (b) Describe how a spool valve controls the direction of flow in hydraulic circuits. (3)
- (c) Explain how PWM is used to control the speed of a DC motor. (3)

- IV. (a) Describe PD control action. What is its advantage over proportional control? (4)
- (b) The rotational velocity ω of the satellite shown in figure is adjusted by changing the length of the beam L . The transfer function between $\omega(s)$ and

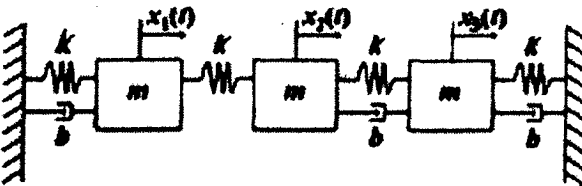
the incremental change in beam length $\Delta L(s)$ is $\frac{\omega(s)}{\Delta L(s)} = \frac{2.5(s+2)}{(s+5)(s+1)^2}$. The

beam length change is $\Delta L(s) = 1/(4s)$. Determine the response of the velocity $\omega(t)$.

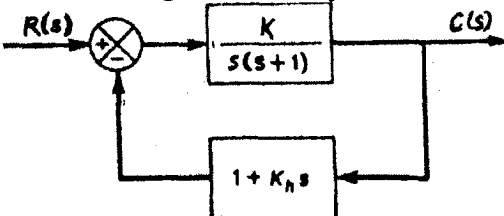


OR

- V. (a) Determine the force-voltage analogous circuit for the system shown in the figure (5)



- (b) For the system shown in figure, determine the values of gain K and constant K_h so that maximum overshoot is 0.2 and peak time is 1s. Find the rise time and 2% settling time of the system. (5)



(Contd...3.)

- VI. Plot the bode diagram for the following loop transfer function and determine the frequency when $20 \log|GH|$ is very close to 0 dB. (10)

$$G(s)H(s) = \frac{12(s + 0.5)}{s^2 + 13s + 30}.$$

OR

- VII. Sketch root locus for the single-loop feedback control system having characteristic equation. (10)

$$1 + G(s)H(s) = 1 + \frac{K(s + 1)}{s(s + 2)(s + 4)^2} = 0.$$

- VIII. (a) A water pump for filling a tank is started using a spring-return push-button switch. When a sensor detects that the tank is full, the pump is stopped. Then the water is heated for 300 seconds using a heater. Draw a ladder diagram to represent this. (4)
- (b) Explain the working of a micro-processor with a neat sketch of its internal architecture. (6)

OR

- IX. (a) A signal to start moving a lift is produced when the lift door is closed and a button has been pressed to select the floor. Draw the truth table and represent the case using suitable logic gate/s. (4)
- (b) Draw and explain the design for a robotic arm for pick and place application, where each joint is actuated by separate pneumatic cylinders. (6)